GIOVANNI QUINONES-VALDEZ

giovas@ucla.edu | 352-256-4917 | https://giovanniquinones.github.io

EXPERIENCE

- **Postdoctoral Fellow |** *Department of Integrative Biology & Physiology, UCLA* 01/2022 – Present

  I develop statistical and computational methods to study the function and regulation of RNA nucleotide variants and RNA processing utilizing multiple modalities of high-throughput sequencing data.

- **Collaboratory Fellow |** *Institute for Quantitative & Computational Biosciences, The Collaboratory, UCLA* 01/2022 – Present

  I provided computational research support to UCLA laboratories through consulting, training sessions, and hands-on bioinformatics workshops.

- **Part-time Scientific Consultant |** *Rattan Life Science* 08/2025 – 09/2025

  Delivered hands-on training to staff scientists in the processing and analysis of single-cell RNA-sequencing (scRNA-seq) data analysis using R and artificial intelligence approaches.

- **Undergraduate Research Assistant |** *Lewin Lab, University of Florida* 03/2014 – 05/2015

  Worked on antioxidant gene therapy approaches for retinal degeneration and contributed to two conference abstracts presented at ISCE and IOVS.

EDUCATION

- **University of California, Los Angeles |** Bioengineering (Bioinformatics focus) | Ph.D. 12/2021
  - GPA: 3.745/4.0
  - Advisor: Dr. Xinshu (Grace) Xiao
  - Dissertation: "Function and Regulation of Nucleotide Variants in RNA"
- **University of Florida |** Biological Engineering | B.S. 05/2015
  - GPA: 3.84/4.0 (Summa Cum Laude)
  - Honors Program
  - Minor: Biomolecular Engineering

PUBLICATIONS

1. King AJ, Amoah K, Zhang L, Bahn JH, Barney RM, **Quinones-Valdez G**, Xiao X. *Allele-specific splicing in the human brain reveals genetic variants linked to protein isoforms and Alzheimer's disease.* Genome Research. https://doi.org/10.1101/gr.281117.125. (2026)
2. **Quinones-Valdez G**, Chang JW, Magaki SD, Vinters HV, Salamon N, Liau L, Wang AC, Fallah A, Owens GC. *Immune landscape of the affected brain in Rasmussen encephalitis.* Scientific Reports. https://doi.org/10.1038/s41598-026-51295-3. (2026).
3. **Quinones-Valdez G**, Amoah K, Xiao X. *Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing*. Nature Communications. https://doi.org/10.1038/s41467-025-64605-6. (2025)

4. Hervoso JL, Amoah K, Dodson J, Choudhury M, Bhattacharya A, **Quinones-Valdez G**, Pasaniuc B, Xiao X. *Splicing-specific transcriptome-wide association uncovers genetic mechanisms for schizophrenia*. The American Journal of Human Genetics. https://doi.org/10.1016/j.ajhg.2024.06.001. (2024)
5. Zheng R, Dunlap M, Lyu J, Gonzalez-Figueroa C, Bobkov G, Harvey SE, Chan TC, **Quinones-Valdez G**, Choudhury M, Vuong A, Flynn RA, Chang HY, Xiao X, Cheng C. *LINE-associated cryptic splicing induces dsRNA-mediated interferon response and tumor immunity*. Molecular Cell. https://doi.org/10.1016/j.molcel.2024.05.004. (2024)
6. Liu Z, **Quinones-Valdez G**, Fu T, Choudhury M, Reese F, Mortazavi A, Xiao X*. L-GIREMI uncovers RNA editing sites in long-read RNA-seq*. Genome Biology. https://doi.org/10.1186/s13059-023-03012-w. (2023)
7. **Quinones-Valdez G**, Fu T, Chan WT, Xiao X. *scAllele: a versatile tool for the detection and analysis of variants in scRNA-seq.* Science Advances. https://doi.org/10.1126/sciadv.abn6398. (2022)
8. Chan TC, Fu T, Bahn JH, Jun HI, Lee JH, **Quinones-Valdez G**, Cheng C, Xiao X. *RNA editing in cancer impacts mRNA abundance in immune response pathways.* Genome Biology. https://doi.org/10.1186/s13059-020-02171-4. (2020)
9. **ENCODE Consortium.** *Perspectives on ENCODE*. Nature. https://doi.org/10.1038/s41586-020-2449-8. (2020)
10. **ENCODE Consortium**. *ENCODE Phase III: Building an Encyclopedia of Regulatory Elements for Human and Mouse*. Nature. https://doi.org/10.1038/s41586-020-2493-4. (2020)
11. **Quinones-Valdez G**, Tran S, Jun HI, Bahn JH, Yang EW, Zhan L, Brummer A, Wei X, VanNostrand E, Pratt G, Yeo GW, Graveley BR, Xiao X. *Regulation of RNA editing by RNA Binding Proteins in Human Cells.* Communications Biology. https://doi.org/10.1038/s42003-018-0271-8. (2019)
12. Yang EW, Bahn JH, Hsiao YH, Tan BX, Sun Y, Fu T, Zhou B, Van Nostrand EL, Pratt, Freese P, Wei X, **Quinones-Valdez G**, Urban AE, Graveley BR, Burge CB, Yeo GW, Xiao X. *Allele-specific binding of RNA Binding Proteins Reveal Functional Genetic Variants in the RNA*. Nature Communications. https://doi.org/10.1038/s41467-019-09292-w. (2019)
13. Hsiao YH, Bahn JH, Yang Y, Lin X, Tran S, Yang EW, **Quinones-Valdez G**, Xiao X. *RNA editing in nascent RNA affects pre-mRNA splicing.* Genome Research. https://doi.org/10.1101/gr.231209.117. (2018)

ARTICLES UNDER REVIEW

1. Zheng R*, **Quinones-Valdez G***, Xiao X, Cheng C. *CryEx.v2: An enhanced protocol for the quantitative detection and visualization of cryptic splicing using transcriptome data.* STAR Protocols
2. Barney RM, **Quinones-Valdez G**, King AJ, Amoah K, Wang W, Xiao X. *Allele-specific alternative polyadenylation links noncoding genetic variation to Alzheimer's disease risk*. Genome Biology.
3. Vavilina A, Calvanese V, Gu Y, Ma F, **Quinones-Valdez G**, Perrod C, Colombo G, Fares I, Ko MW, Kaaja I, Julia, Aguade-Gorgorio J, Nadel B, Liebscher S, Wang Y, Goodridge H, Crooks GM, Pellegrini M, Xiao X, Schenke-Layland K, Mikkola HKA. *Developmentally regulated long and short isoforms of MLLT3 balance human hematopoietic stem cell fate decisions.* Cell Stem Cell.

CONFERENCE ABSTRACTS

1. Kaaja I, Wang Y, Kilpivaara O, Wartiovaara-Kautto U, **Quinones-Valdez G**, Gu Y, Turmon A, Ma F, Wang Y, Aguade-Gorgorio J, Ko MW, Grimbert S, Wang Y, Katainen L, Park S, Pellegrini M, Xiao X, Seet CS, Chang VY, Arboleda V, Backus K, Mikkola HKA. *Tracking the Stemness Mechanisms in*

Refractory Mecom Rearranged Acute Myeloid Leukemias at Single Cell Resolution. European Hematology Association. 2025.

2. Frenz-Wiessner S, Wang Y, Rohde Y, **Quinones-Valdez G**, Issa H, Gonçalves-Dias J, Schuschel K, Mack P, Ko MW, Wang Y, Ma F, Sakamoto K, Chang V, Xiao X, Klusmann JH, Mikkola HKA. *Single-cell tracing of myeloid leukemia evolution in down syndrome reveals a latent progenitor bridging disease onset, progression and relapse*. Blood, 2025

3. Biswal MR, Ildenfonso CJ, Rossmiller BP, Mao H, Li H, Zhu P, Tong Y. **Quinones-Valdez G**, Lewin AS. *Delaying Retinal Degeneration in a Mouse Model of Geographic Atrophy: An Antioxidant Gene Therapy Approach*. Investigative Ophthalmology and Visual Science. 2015.

4. Biswal MR, Ildefonso CJ, Wang Z, Mao H, Li H, Zhu P, **Quinones-Valdez G**, Lewin AS. *Effectiveness of Antioxidant Gene Therapy in Delaying Retinal Degeneration*. Molecular Therapy. 2015.

TEACHING AND MENTORING

- **Instructor for the workshop "Unix Command Line"** — 01/2026 – 04/2026 (Quarterly)
  Institute for Quantitative and Computational Biosciences, UCLA
- **Instructor for the workshop "Single Cell RNA-Seq. Data Analysis with R"** — 01/2022 – 03/2025 (Quarterly)
  Institute for Quantitative and Computational Biosciences, UCLA
- **Instructor for the Bruins in Genomics (BIG) summer program** — 07/2024, 07/2023, 07/2020
  Institute for Quantitative and Computational Biosciences, UCLA
- **Guest lecturer for the course "Intro to High-Throughput Data Analysis"** — 03/2024 – 06/2024, 03/2023 – 06/2023
  Department of Biomathematics, UCLA
- **Assistant instructor for the course "Concepts in Molecular Biosciences"** — 01/2024 – 03/2024
  Department of Molecular Biology, UCLA
- **Teaching Assistant for the course "Molecular Parasitology"** — 09/2017 – 12/2017
  Department of Microbiology, Immunology, and Molecular Genetics, UCLA
- **Member of the Xiao Lab at UCLA** — 01/2016 – Present
  - Mentored junior graduate and undergraduate students.
  - Interviewed and hired students to aid with computational analysis.
  - Lead a group of three undergraduate students to carry out a research project on allele-specific RNA processing dysregulation in Alzheimer's Disease.

AWARDS

- **ASHG Trainee Research Excellence Award winner** — 11/2024
  Awarded by the American Society of Human Genetics (ASHG) for outstanding trainee research presentation and scientific contribution at the annual meeting
- **QCBio Collaboratory Postdoctoral Fellowship** — 02/2022 – 03/2025
  Prestigious funding mechanism available to postdocs and project scientists conducting research in computational biology and bioinformatics.
- **Young Bolivian Abroad Award** — 12/2018
  Recipient of the tarijeno abroad award (diaspora category), awarded by the regional government of Tarija, in recognition of academic achievement and contributions as a representative of Tarija abroad.
- **UCLA Bioengineering Supplementary Fellowship** — 03/2017
  Fellowship support awarded by the UCLA department of bioengineering to support

graduate research and training.

- **UCLA Bioengineering Research Fellowship** 01/2016
  Competitive graduate fellowship supporting research in bioengineering
- **Davis Scholarship for Undergraduate Education** 08/2011 – 05/2015
  Merit-based scholarship awarded through the Davis United World College Scholars Program, supporting graduates of the United World Colleges network.

RESEARCH TALKS

- **Title: "Long-read RNA-seq demarcates cis- and trans-directed alternative RNA splicing"**
  - American Society of Human Genetics, *Featured Plenary Talk* 11/2024
  - UCLA Riboforum, *Selected Talk* 01/2024
  - American Society of Human Genetics, *Poster presentation* 11/2023
  - International Society for Computational Biology, *Selected Talk* 11/2023
  - UCLA QCBio Retreat, *Selected Talk* 09/2023
- **Title: "scAllele, a versatile tool for the detection and analysis of variants scRNA-seq"**
  - Open Box Science, *Invited Talk* 04/2023
  - Southern California Systems Biology Consortium, *Poster presentation* 04/2022
  - ENCODE Consortium Retreat, *Talk* 03/2022
  - UCLA QCBIO Seminar, *Poster presentation* 10/2021
- **Title: "Regulation of RNA editing by RNA-binding proteins in human cells"**
  - UCLA QCBIO Retreat, *Poster presentation* 09/2018
  - ENCODE Consortium Retreat, *Poster presentation* 02/2018

JOURNAL REVIEWER

- Genome Biology 03/2026
  11/2025
  05/2026
- BMC Genomics 01/2026
- Genomics, Proteomics, and Bioinformatics 08/2024
- Communications Biology 03/2023

SOFTWARE DEVELOPMENT

- **isoLASER –** Lead Developer 10/2025
  Built in Python | GitHub: https://github.com/gxiaolab/isoLASER, PyPi
  isoLASER performs variant calling, haplotyping, and allele-specific splicing analysis in long-read RNA-Sequencing data. This one-stop solution enables the classification of genetically regulated splicing events revealing transcriptome-wide patterns of regulation and disease-relevant isoform regulation such as the HLA gene family and Alzheimer's disease risk genes.
- **scAllele –** Lead Developer 09/2022
  Built in Python | GitHub: https://github.com/gxiaolab/scAllele, PyPi
  scAllele is an integrative tool that performs high-confidence calls of small variants in single-cell RNA-Sequencing using a graph-theory approach. Additionally, scAllele enables allele-specific splicing analysis, a unique feature not previously used at the single-cell level.
- **L-GIREMI –** contributor 07/2023
  Built in Python | GitHub: https://github.com/gxiaolab/L-GIREMI

L_GIREMI uses a robust statistical approach based on mutual information for the detection of A-to-I RNA editing in long-read RNA sequencing data unveiling insights about the distribution of these sites on individual RNA molecules.

- **BEAPR –** contributor 03/2019
  Built in C++, R | GitHub: https://github.com/gxiaolab/BEAPR
  BEAPR identifies allele-specific binding events in eCLIP-Seq data to functionally characterize genetic variants. Taking into account crosslinking-induced sequence propensity and variations between replicated experiments, BEAPR achieves high accuracy calls and identifies clinically relevant variants.

COMMUNITY ENGAGEMENT AND SERVICE

- **Diversity and Science Lectures (DASL)** 06/2024 – 08/2024
  Member of the DASL organization at UCLA that aims to promote diversity and inclusion through scientific seminars, social activities, and workshops.
- **Reviewer for the United World Colleges (UWC)** 10/2024
  Reviewed applications for the Bolivian UWC selection committee, contributing to 10/2023
  the evaluation and selection of scholarship recipients. 10/2022
- **UCLA Latino Alumni Association** 10/2022 – 06/2023
  Guided undergraduate Latino students in strategizing their paths toward graduate school or securing employment opportunities.
- **Dashew Center for International Students and Scholars** 09/2016 – 06/2017
  Assisted incoming international students to help them adapt to the university and life in the United States
- **Engineering Graduate Student Association (eGSA)** 09/2015 – 12/2021
- **Health-Oriented Professional Engineering Students (HOPES)** 09/2014 – 05/2015
- **Society of Hispanic Professional Engineers (SHPE)** 09/2012 – 05/2015

SKILLS

- **Computational skills**
  - Analysis of Next Generation Sequencing (NGS) data, including bulk RNA-seq, single-cell RNA-Seq, and Third Generation Sequencing data (PacBio, Nanopore)
  - Software development using Python
  - Software packaging and distribution (PyPI, Apptainer, GitHub)
  - Machine Learning in Genomics using Scikit-Learn
  - Programming languages: proficient with Python, Bash, and R
  - Carried out projects on Google Cloud Platform (GCP)
  - Operating Systems: Linux/Unix/Ubuntu, Mac OS, Windows
- **Laboratory skills**
  - PCR
  - RNA and Protein expression analysis
  - Cell culture
  - Optical Coherent Tomography (OCT) and Electroretinogram (ER)
- **Languages**
  - Native or bilingual language proficiency: Spanish and English.
  - Limited working proficiency: French and Italian